

[illegible]

DS-56

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TOTAL NUMBER OF INTERVIEWS

Total Interview counts: 25

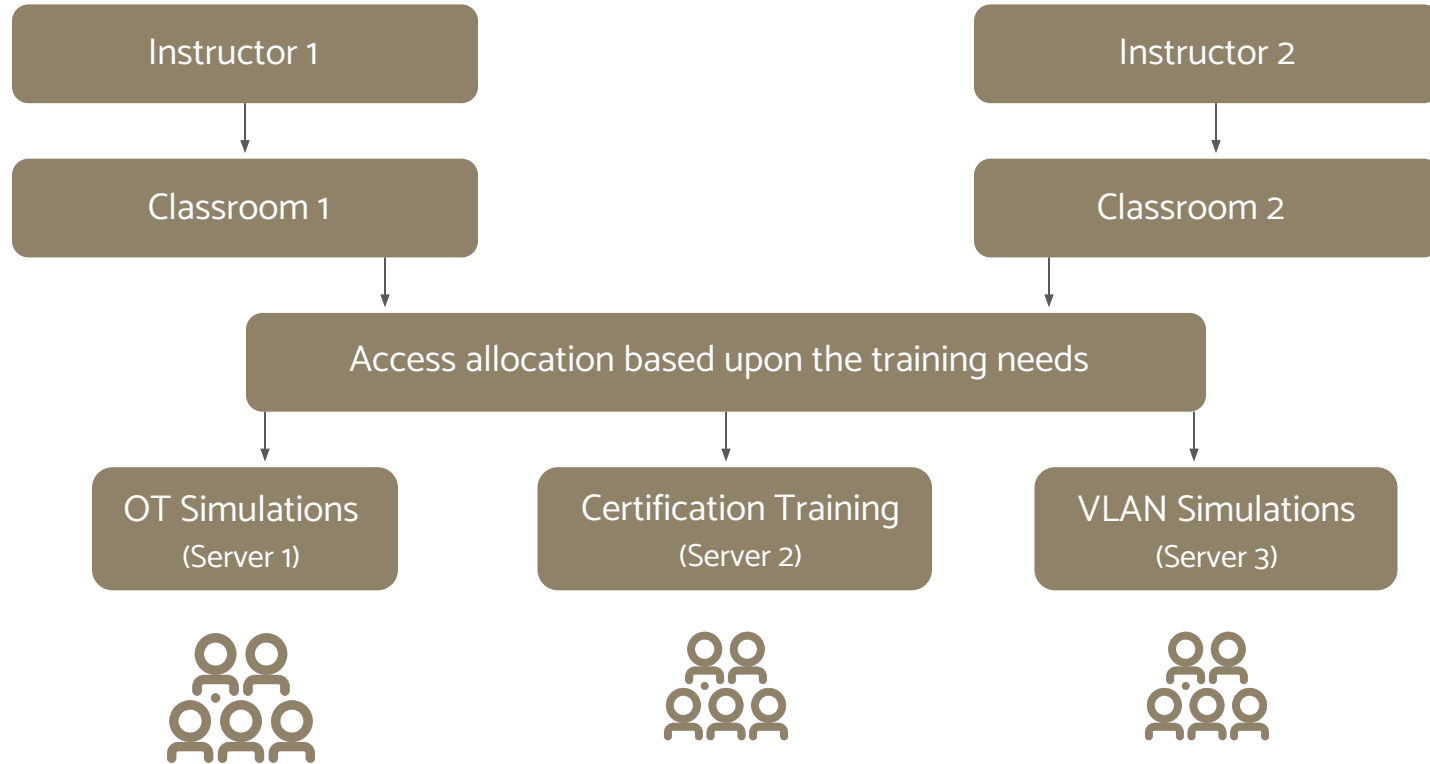
PROBLEM STATEMENT

There is a need for a “virtual platform” to provide better training for students and instructors at the DoS.



FLOW DIAGRAM FOR TESTING

From a Non-Technical Point of View



N number of students can access these training simultaneously

Beneficiary Discovery

HYPOTHESIS

What did you think beginning the week?

Why do they want VLAN training?

EXPERIMENTS

What did you do this week?

Had a meeting with our Project Sponsor (Phillip) to understand more on the OT trainings and its security aspect as well as understanding VLAN training in depth.

RESULTS

What did you discover?

VLAN training will involve a separate secure network system that mimics the DoS current security network. (i.e. Honeypots for Security purpose).

OT training (Optional), main goal is to provide training on Physical OT devices.

ACTIONS

What's next? How do you proceed?

We need to find the softwares to give that hands-on training for Building Management systems (BMS) for OT training.

The Mission Model Canvas

Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest (Instructor),
Ben Hansen (Our OT guy)

Key Activities

- Making a virtual platform to integrate different environments.
- Keeping it secure.

Value Proposition

1. To audit OT systems.
2. Sandbox style training on various environments.
3. (OT, Networks)
4. Get real-time monitoring and feedback.
5. Hands-on security training through certificates.

Buy-in & Support

- Leadership Support
- Financial & Technical Resources
- Stakeholder & Compliance Elements.

Deployment

- Implementation Phases.
- Technical Infrastructure
- User Adoption Strategy

Mission Budget

\$500,000 allocated for building the entire Virtual Network System.

Achievement

Providing a way for hands-on training in a virtual setting.

Beneficiaries

1. OT security personnel (students)
2. Engineers working in Springfield, VA (students)
3. Instructors at the DoS
4. People with no cybersecurity background (Students)

Key Resources

- Technical Infrastructure (like Physical servers and VMs)
- Software Resources (like 3rd vendor softwares for OT environments)
- Knowledge Resources (Content and materials)

VALUE PROPOSITION CANVAS - (PHILIP, INSTRUCTOR)

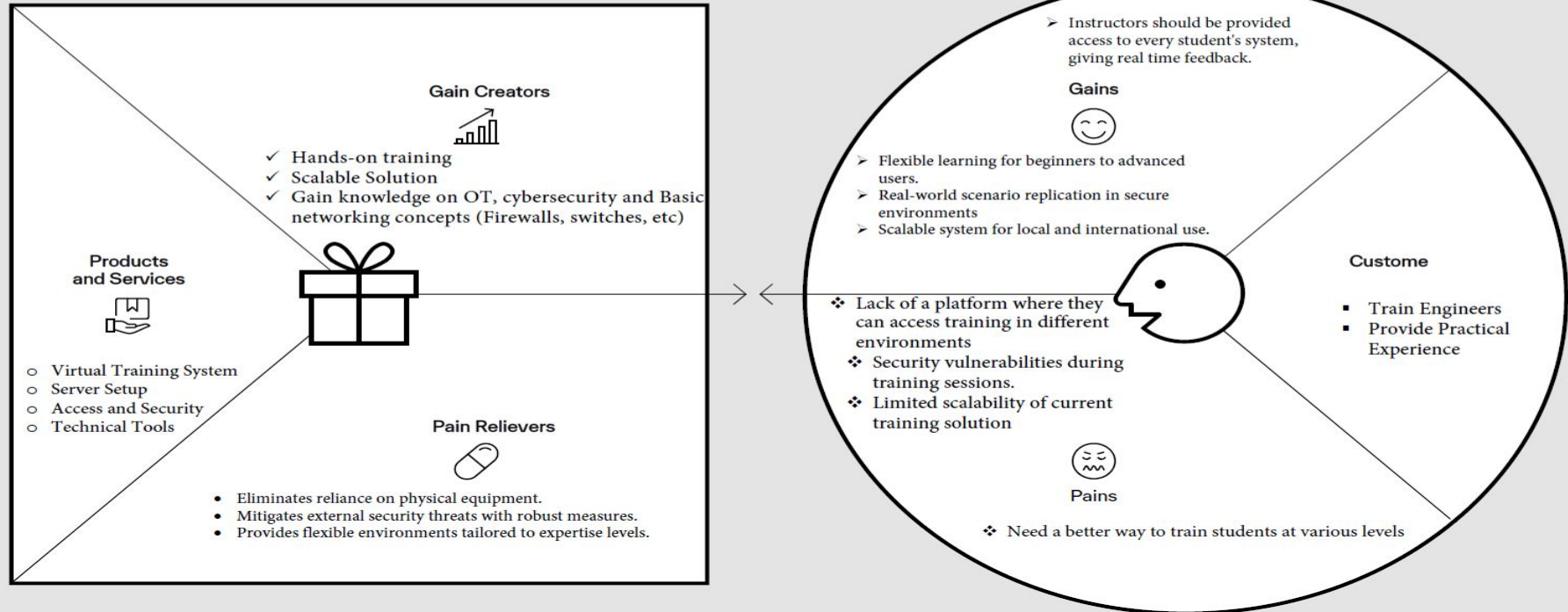
Phillip (Project Sponsor)

The Value Proposition Canvas

Value Proposition:

Flexible, scalable, and secure virtual training platform providing hands-on experience in cybersecurity, OT systems, and IT concepts.

Customer Segment: Engineers in Springfield, Virginia (Department of State) and potential global users (U.S. embassies) seeking practical training in OT security and IT certifications.



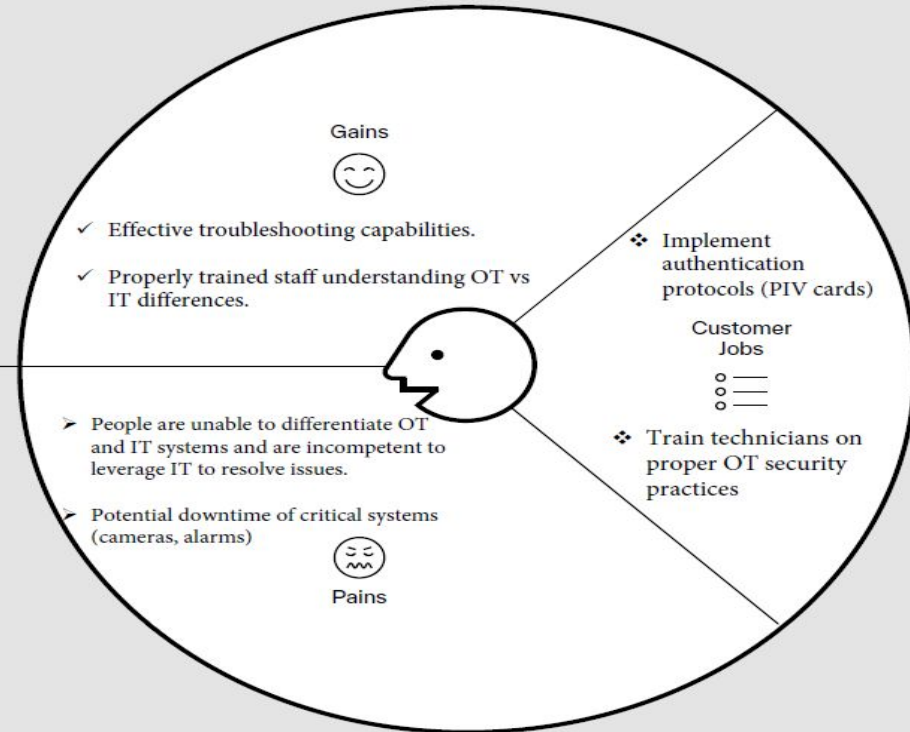
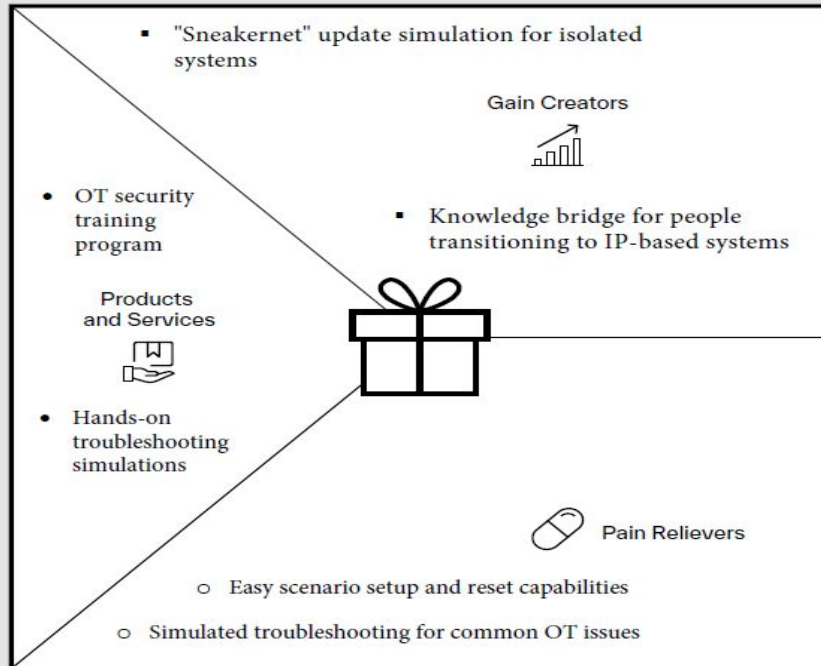
VALUE PROPOSITION CANVAS - (BEN HANSEN)

Ben Hansen

The Value Proposition Canvas

Value Proposition: Specialized OT Security Training Program for DoS Risk Mitigation

Customer Segment: Risk Mitigation Unit and OT Security Personnel at Department of State



WEEK 11 MVP + EXPERIMENTS CONDUCTED

MVP Experiment Canvas

Project name:

A virtual platform delivering hands-on training across different environments.

Made by:

Ananmay Rajan

4. Customer Engagement ❤️

How are you going to engage the customers who used your MVP, to learn from their experience and gather feedback to improve your Value Proposition?

Follow-up email / call

1. Your Customer Segment 👤

Which segment of your target customers will experience the most value from your MVP, and can you learn from them?

2. Value Proposition 📄

My company, __ (insert of company) __, is developing __ (a defined offering) __ to help __ (a defined audience) __ (solve a problem) __ with __ (secret sauce/differentiation) __.

3. Channel(s) 🗣️

How are you going to reach your Customer Segment and "deliver" your Value Proposition to them?

Virtual platform with secure access

5. Riskiest Assumption(s) 🎯

What you are setting out to test with your MVP?

Are these assumptions about the Problem, Solution or Implementation?

Insufficient data sanitization measures resulting in residual data exposure between training sessions.

6. Experiment Format 🧪

WHAT type of experiment are you going to run?

- Simulated virtual training labs managed manually.
- Goal: provide a Windows-based user interface.
- Test with small groups of OT personnel and other staff @DoS in Springfield, VA.

7. Scenario / Workflow 🔄

HOW does that work? Describe the steps from beginning to end.

Secure Login: Users access the platform securely.

Task Assignment: Instructors assign tasks (e.g., simulations, team exercises).

Real-Time Monitoring: Instructors monitor activities and provide feedback.

Performance Logging: System logs user performance and feedback.

Temporary VM Storage: Servers store VM data temporarily for review, then erase securely.

8. Metrics 📊

What will you measure? (Can be multiple metrics)

- Multiple User Access.
- Feedback scores on how well the platform works.

9. Success Criteria 🏆

How do you qualify and/or quantify a successful outcome of this experiment?

At least 60 users complete the training with positive feedback

10. Results (Expected) 📈

What are the qualitative and / or quantitative

- Train 20 users, then expand to 60.
- Gather usability and effectiveness data.
- Simulate Red vs Blue Team scenarios and measure outcomes.

11. Learnings & Insights 🧠

What are your key learnings?

What are your key surprises?

Did you get enough results?

What do they tell you about your Riskiest Assumption(s)?

Assess if virtual training meets the needs of technical and non-technical personnel.

VALIDATED

INVALIDATED

INCONCLUSIVE

12. Next steps 🚀

What do the results tell you to do next?

Are you going to Pivot, Pursue, or Stop?

See if the platform is scalable, flexible and robust for simultaneous use

MVP (Contd..) (Hardware Requirements for now)

128 vCPUs, 768 GB RAM, and
NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

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NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

Use Case

- ✓ Simulating real-world cyber-attacks and training on OT systems.
- ✓ Running multiple simulations simultaneously.
- ✓ Instructor has access to multiple VM's through Hypervisor.

For each VM

CPU: 4 vCPUs (For running multiple simulation processes)

RAM: 8-12 GB (for advanced security simulations)

Storage: 100-150 GB SSD/NVMe (if storing logs or running databases)

Network: 1 Gbps or 10 Gbps virtual NIC (for high-speed data exchange)



Virtual Machines

(Physical Servers for OT training)